

OSSP PRACTICAL PROGRAM 3: -

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Code: -

```
#include
<stdio.h>

#include
<stdlib.h>

#include
<unistd.h>

#include
<sys/types.h>

#include
<sys/wait.h> int
main() {

    pid_t pid;

    printf("Parent process
started\n"); printf("Parent
```

```
PID : %d\n", getpid());

printf("Parent PPID : %d\n\n",
getppid()); pid = fork();

if (pid < 0) {

    perror("fork
    failed"); return 1;

}

if (pid == 0) {

    printf("----- Child Process ---\n");

    printf("Child PID : %d\n",
    getpid()); printf("Child
    PPID : %d\n", getppid());
    printf("Child is running...\n");

    sleep(5);

    printf("Child process
    terminated.\n"); exit(0);

}

else {
```

```

printf("----- Parent Process ---\n");

printf("Parent PID : %d\n",
getpid()); printf("Child
PID : %d\n", pid);

printf("Parent is waiting for
child.....\n");

wait(NULL);
printf("Child process has
terminated.\n"); printf("Parent
process terminated.\n");

}
return 0;
}

```

OUTPUT: -

```
practical — nano practicalq3.c — 80x24
UW PICO 5.09 File: practicalq3.c

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>

int main() {

    pid_t pid;

    printf("Parent process started\n");

    printf("Parent PID : %d\n", getpid());

    printf("Parent PPID : %d\n\n", getppid());

    ^G Get Help  ^O WriteOut  ^R Read File  ^Y Prev Pg   ^K Cut Text   ^C Cur Pos
    ^X Exit      ^J Justify    ^W Where is   ^V Next Pg   ^U UnCut Text ^T To Spell

apple@apples-MacBook-Pro-2 practical % nano practicalq3
apple@apples-MacBook-Pro-2 practical % nano pratical q3.c
apple@apples-MacBook-Pro-2 practical % nano practical.c
apple@apples-MacBook-Pro-2 practical % nano practicalq3.c
apple@apples-MacBook-Pro-2 practical % gcc practicalq3.c -o practicq3
apple@apples-MacBook-Pro-2 practical % ./practicq3
Parent process started
Parent PID : 3125
Parent PPID : 2807

----- Parent Process -----
Parent PID : 3125
Child PID : 3126
Parent is waiting for child...
----- Child Process -----
Child PID : 3126
Child PPID : 3125
Child is running...
Child process terminated.
Child process has terminated.
Parent process terminated.
```